CHIARA GARDENGHI

chiaragardenghi@rochester.edu chiaragardenghi.com

APPOINTMENTS

University of Rochester, Simon Business School
Postdoctoral Associate in Economics. Supervisors: Jeanine Miklós-Thal, Ellie Prager 2025–present

EDUCATION

New York University, Stern School of Business
Ph.D. in Economics. Committee: M. Dickstein, C. Conlon, L. Cabral, G. Brancaccio 2019–2025

Bocconi University
M.Sc. in Economics (ESS), *summa cum laude*. Main advisor: G. Tabellini 2016–2018

Bocconi University
B.Sc. in Economics (CLES), *summa cum laude*. Main advisor: S. Nunnari 2013–2016

Singapore Management University
Spring Exchange Program 2016

London School of Economics and Political Science
Summer Exchange Program 2015

REFERENCES

Prof. Michael J. Dickstein
Dept. of Economics
NYU Stern School of Business
✉ michael.dickstein@nyu.edu

Prof. Christopher Conlon
Dept. of Economics
NYU Stern School of Business
✉ cconlon@stern.nyu.edu

Prof. Luís Cabral
Dept. of Economics
NYU Stern School of Business
✉ luis.cabral@nyu.edu

RESEARCH FIELDS

Industrial Organization, Health Economics, Innovation, Competition Policy

WORKING PAPERS

"Bundled Loyalty Discounts in Healthcare: Evidence from Physician-Administered Vaccines" (JMP)
Winner of the 2025 Young Economists' Paper Award at the EARIE Annual Conference

In many industries, multi-product sellers offer bundled loyalty discounts to buyers who refrain from purchasing competing products from rivals. These contracts can increase sales volumes by helping sellers establish a strong customer base across their product portfolios, but they also risk foreclosing competing sellers, even those offering superior products. I examine the welfare implications of bundled loyalty discounts in the context of physician-administered vaccines, which medical providers purchase and administer directly to their patients. I use prior antitrust cases involving vaccines to recover the proprietary terms of bundled discount contracts. I then develop a model of medical providers' demand and pharmaceutical companies' supply of vaccines. Using all-payer claims data from Colorado to estimate the model, I recover medical providers' preferences, vaccines' marginal costs, and the magnitude of bundled discounts that rationalizes providers' observed purchasing patterns. I use the model to evaluate the equilibrium and welfare effects of banning bundled discounts, allowing sellers and buyers to re-optimize their choices. I show that, in the absence of discounts, providers purchase fewer and lower-quality products, particularly at the expense of optional vaccines not required for school access. Insurers benefit in the short term through lower immunization coverage costs. The effect on pharmaceutical companies varies, depending on the scope and composition of their vaccine portfolios.

"Public and Private Innovation: Complements or Substitutes?"
With Luís Cabral, Deepak Hegde, and Junrui Lin

We develop a two-player multi-period game in which a public innovator and a private innovator allocate effort across research and development over time. We show that, when decisions are simultaneous, innovation efforts are strategic substitutes, as each player's effort reduces the marginal value of the other's. When decisions are sequential, however, innovation efforts are strategic complements, as knowledge accumulated through spillovers reduces future innovation cost. We test these predictions in the context of U.S. biomedical research over the 2000-2019 period. Leveraging the granting of FDA Accelerated Approval status as a positive shock to private clinical trial activity, we find that the public sector reduces research funding allocated to affected disease areas in the short run—consistent with substitutability—but increases it in the medium-to-long run—consistent with complementarity. A symmetric pattern emerges when exploiting federal initiatives and executive actions as positive shocks to public funding: private clinical trial activity declines in the short run before expanding in the medium-to-long run. Taken together, our findings suggest that substitutability and complementarity are not competing characterizations but rather two faces of strategic interaction.

"Healthcare Innovation Across the Racial Divide: The Interplay of Public and Private Financing"
With Luís Cabral, Deepak Hegde, and Junrui Lin

U.S. pharmaceutical companies invest nearly $100 billion annually in R&D, while the National Institutes of Health (NIH) allocates around $50 billion to biomedical research. Experts argue these investments should reflect public health needs and scientific opportunity, yet systematic evidence on allocation patterns remains limited. Using machine learning, we compile a novel panel dataset linking NIH funding and industry clinical trials across major diseases over two decades to disease-specific characteristics: burden, scientific opportunity, revenue potential, and the race and gender composition of affected populations. We find that both sectors invest more in high-burden diseases, but private firms also prioritize high-revenue conditions. Strikingly, the private sector underinvests in diseases that disproportionately affect Black Americans—even after adjusting for burden, revenue, and opportunity. NIH funding partially offsets this gap but compensates for only about 80% of the "missing" trials for conditions affecting Black Americans. We identify disease areas where gaps remain largest, offering a roadmap for targeted policy action. Our findings highlight NIH's essential—but incomplete—role in correcting racial disparities in biomedical innovation and show how recent cuts to NIH and Medicaid risk deepening these inequities.

"Supply-Side Incentives for Medical Technology Adoption"

I provide novel evidence that physicians respond to changes in insurance reimbursement for in-office treatments, although the direction and potential spillover effects of these responses may vary. To estimate physician elasticity to private insurance reimbursement, I exploit a plausibly exogenous regulatory change in Medicare payment. Specifically, I document how private reimbursement typically follows Medicare reimbursement (Clemens and Gottlieb, 2016; Chan and Dickstein, 2019) and use the estimated payer-specific relationship between the two prices to construct a valid instrument. My findings show that supply generally exhibits a positive slope—lower reimbursement reduces treatment provision. However, for certain discretionary treatments, the supply curve is negatively sloped. I provide suggestive evidence that physicians compensate for reduced reimbursement by increasing provision of these and related treatments that can be easily combined in single patient visits.

Work in Progress

"Antitrust Remedies and Complementarities: Evidence from Pharmaceuticals"

"Cost-Plus Reimbursement Contracts and Market Segmentation"
With Jinglin Wang

Teaching

Graduate Level (NYU Stern):

Industrial Organization, TF for Prof. Chris Conlon, Luís Cabral, Giulia Brancaccio — Spring 2023, 2025

PhD Math Camp, Instructor — Summer 2024

MBA/EMBA Level (NYU Stern):

Healthcare Markets, TF for Prof. Michael J. Dickstein — Spring 2022

Undergraduate Level (NYU Stern and Bocconi):

Health Economics, TF for Prof. Michael J. Dickstein — Spring 2022

Microeconomics with Algebra, TF for Prof. Luís Cabral — Fall 2021, 2022

Microeconomics, TF for Prof. Maristella Botticini — Fall 2018

Research Experience and Other Employment

Research Assistant for Prof. Guido Tabellini, Bocconi University — 2018–2019

Intern at The Boston Consulting Group (Risk&Banking Team), Milan — 2017

Visiting Researcher at IGIER, Bocconi University — 2017–2019

Visiting Student at Nunnari Lab for Economic Experiments, Bocconi University — 2016–2017

Honors and Awards

Young Economists' Paper Award, EARIE Annual Conference — 2025

Harold W. MacDowell Prize, awarded to "the PhD graduate who best exemplifies the qualities of and dedication to scholarship" — 2025

Center for Sustainable Business research grant ($\$2,500$) — 2024

Center for Global Economy and Business research grant ($\$7,000$) — 2021

Edwin and Diane Elton Fellowship, NYU Stern — 2023–2024

Talks (* : scheduled, D: discussant)

CEPR-JIE Applied IO (D), Luís Cabral's 65th birthday conference at Nova SBE Lisbon, Marketing Science, Markets and Waves (D), EARIE ("Rising Stars") — 2026

Bocconi-EIEF-Bologna Winter IO Workshop, International Industrial Organization Conference (IIOC, "Rising Stars" session), TSE 3rd Health Economics Conference, European Association for Research in Industrial Economics (EARIE, "Rising Stars" session), ASHEcon (D), Cornerstone Research, Analysis Group, Rochester Simon, Indiana Kelley, UOregon, Stockholm School of Economics, Universidad Carlos III de Madrid — 2025

American Society of Health Economists (ASHEcon), NBER Health Economics Research Summer Bootcamp, AEA Workshop for Women in Health Economics and Policy, NYU and NYU Stern Seminars: Industrial Organization, Applied Micro, Econometrics — 2021–2024

Service

Referee: *Journal of Health Economics, Journal of Economics and Management Strategy, Review of Industrial Organization, Quarterly Review of Economics and Finance*

Co-organizer: 3rd Antitrust Economics and Competition Policy Conference (Rochester), Junior Faculty Brownbag (Rochester), Metrics Reading Group (NYU)

Mentor: PhD and MSc in Quantitative Economics students at NYU

Others

Software and Programming: Python, Stata, LaTeX, GitHub, Bash (Linux), SQL

Languages: English (Fluent), Italian (Native), French (Intermediate)

Citizenship: Italian